FINM3013 Introduction to Financial Derivatives

Chapter 10 Exercises

Solutions

1.

The present value of the strike price is $60e^{-0.12\times 4/12} = \$57.65$. The present value of the dividend is $0.80e^{-0.12\times 1/12} = 0.79$. Because

$$5 < 64 - 57.65 - 0.79$$

the condition in equation (10.8) is violated. An arbitrageur should buy the option and short the stock. This generates 64 – 5 = \$59. The arbitrageur invests \$0.79 of this at 12% for one month to pay the dividend of \$0.80 in one month. The remaining \$58.21 is invested for four months at 12%. Regardless of what happens a profit will materialize.

If the stock price declines below \$60 in four months, the arbitrageur loses the \$5 spent on the option but gains on the short position. The arbitrageur shorts when the stock price is \$64, has to pay dividends with a present value of \$0.79, and closes out the short position when the stock price is \$60 or less. Because \$57.65 is the present value of \$60, the short position generates at least 64 – 57.65 – 0.79 = \$5.56 in present value terms. The present value of the arbitrageur's gain is therefore at least 5.56 – 5.00 = \$0.56.

If the stock price is above \$60 at the expiration of the option, the option is exercised. The arbitrageur buys the stock for \$60 in four months and closes out the short position. The present value of the \$60 paid for the stock is \$57.65 and as before the dividend has a present value of \$0.79. The gain from the short position and the exercise of the option is therefore exactly 64 – 57.65 − 0.79 = \$5.56. The arbitrageur's gain in present value terms is 5.56 – 5.00 = \$0.56.

2. If the call is worth \$3, put-call parity shows that the put should be worth

$$3 + 20e^{-0.10\times 3/12} + e^{-0.1\times 1/12} - 19 = 4.50$$

This is greater than \$3. The put is therefore undervalued relative to the call. The correct arbitrage strategy is to buy the put, buy the stock, and short the call. This costs \$19. If the stock price in three months is greater than \$20, the call is exercised. If it is less than \$20, the put is exercised. In either case the arbitrageur sells the stock for \$20 and collects the \$1 dividend in one month. The present value of the gain to the arbitrageur is

$$-3 - 19 + 3 + 20e^{-0.10\times 3/12} + e^{-0.1\times 1/12} = 1.50$$

3. Consider a portfolio that is long one option with strike price K_1, long one option with strike price K_3, and short two options with strike price K_2. The value of the portfolio can be worked out in four different situations

$S_T \le K_1$: Portfolio Value $= 0$

$K_1 < S_T \le K_2$: Portfolio Value $= S_T - K_1$

$K_2 < S_T \le K_3$: Portfolio Value $= S_T - K_1 - 2(S_T - K_2) = K_2 - K_1 - (S_T - K_2) \ge 0$

$S_T > K_3$: Portfolio Value $= S_T - K_1 - 2(S_T - K_2) + S_T - K_3 = K_2 - K_1 - (K_3 - K_2) = 0$

The value is always either positive or zero at the expiration of the option. In the absence of arbitrage possibilities it must be positive or zero today. This means that

$$c_1 + c_3 - 2c_2 \ge 0$$

or

$$c_2 \le 0.5(c_1 + c_3)$$